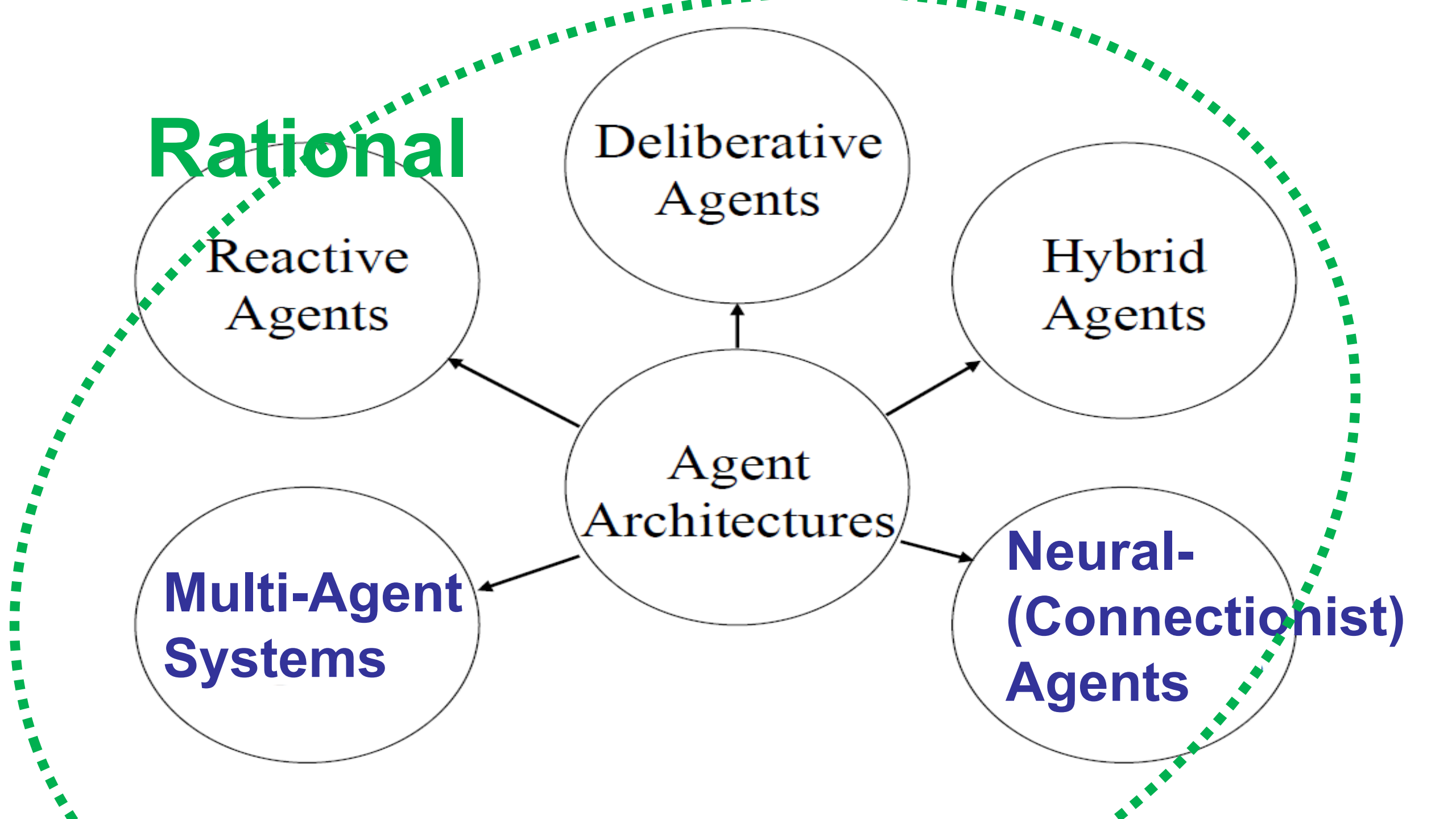


CSC3044S
Artificial Intelligence:
Agent Architectures II

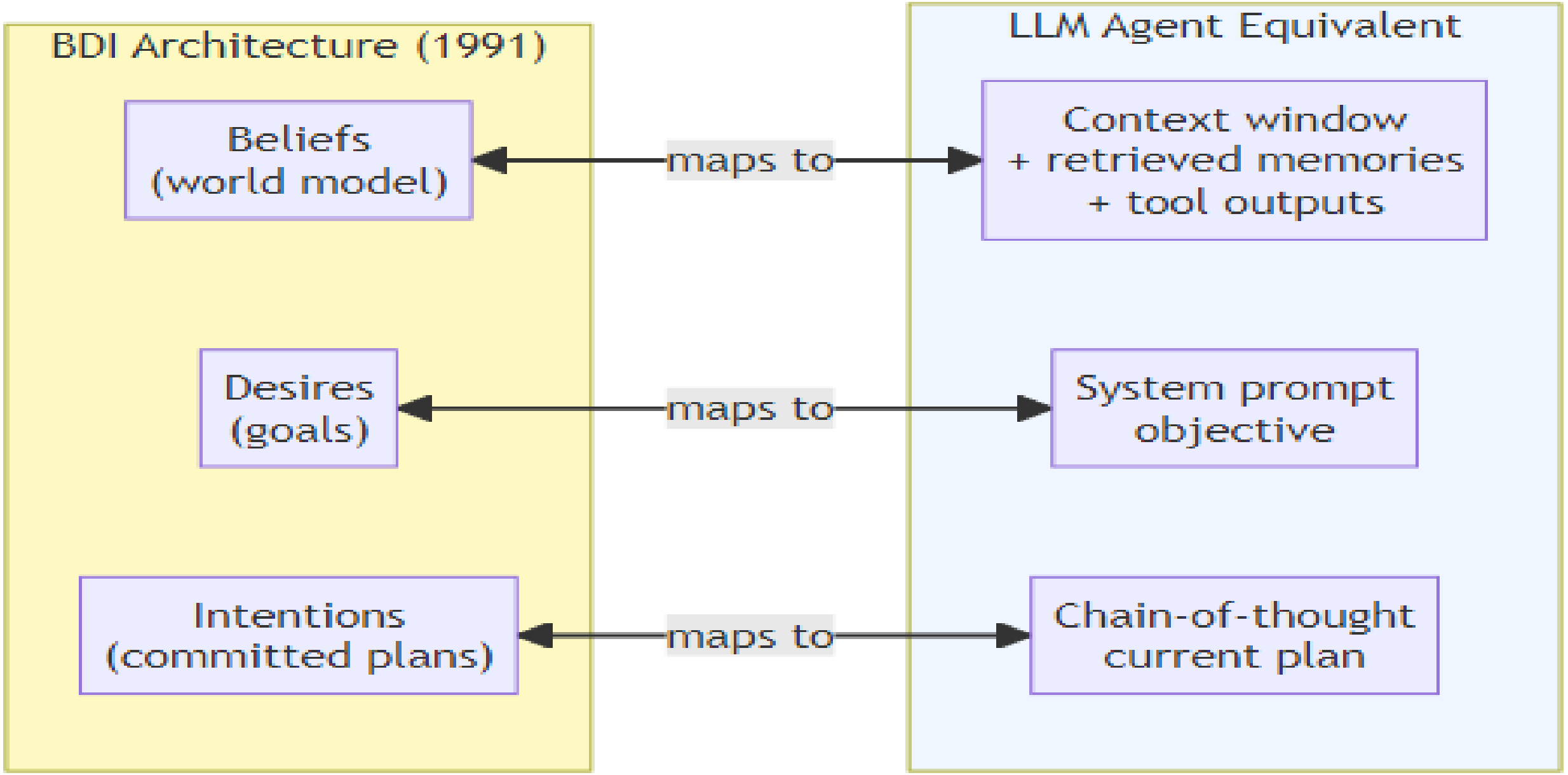
Geoff Nitschke
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Rational
Deliberative Agents
Reactive Agents
Hybrid Agents
Agent Architectures
Multi-Agent Systems
Neural-(Connectionist) Agents

Agent Architecture Examples

- **Reactive:** *Subsumption, Boids, Continuum, Stigmergy.*
- **Deliberative (Symbolic AI):** *Belief, Desires Intention (BDI).*
- **Cognitive (Neural AI):** AASMO, Soar, ACT-R, Brahms, LIDA, PreAct, Cougaar, PRODIGY, FORR.
- **Hybrid:** 3T, AuRA, Brahms, GAIuS, GRL, ICARUS, InteRRaP, TinyCog, *TouringMachines*.
- **What about:** Claude, Gemini, DeepSeek, …?

Deliberative & Hybrid Architectures



What is an Emergent Phenomena?

Emergent Behaviour

- Important but poorly understood phenomenon – Found in behaviour-based, reactive (agent-based) systems and nature.
- Global behaviours “emerge” from interactions between rules and environment (sum greater than the parts).
- *What types of individual (rule, behaviour) interactions lead to what types of (global) behaviours?*



Emergent Behaviour

Emergent Behaviour

- Individual (base) behaviours are (pre)-coded.
- Observed (global) behaviours are emergent.
- There is no one-to-one mapping between the two.
- When observed behaviour “exceeds” programmed behaviour – then we have *emergence*.

Deep-Learning (DL) and Emergence

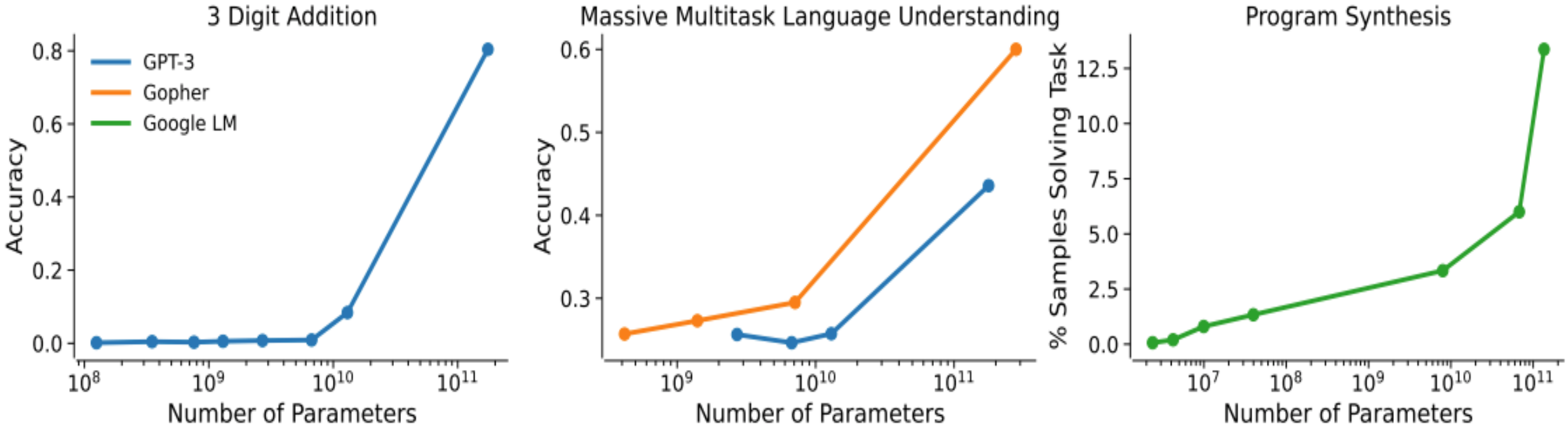
- Emergence is the rule (not the exception) in DL systems:
  - ANN output is emergent (black-box) → Generalisation (Universal function approximation).
  - Only ANN (structure, connections), and training algorithm are designed.



- *What if we increase model parameters → billions (connection weights, activation functions)?*

Emergence in LLMs[1]

- **Is this emergence?** – Model task-performance improves with scale – but what about "all-or-nothing" emergence? (Either a problem-solving behaviour emerges or not).



1. Krakauer, D., et al. (2026). Large language models and emergence: a complex systems perspective. Phil. Trans. R. Soc. A 14 May 2026; 384 (2320): 20250014.

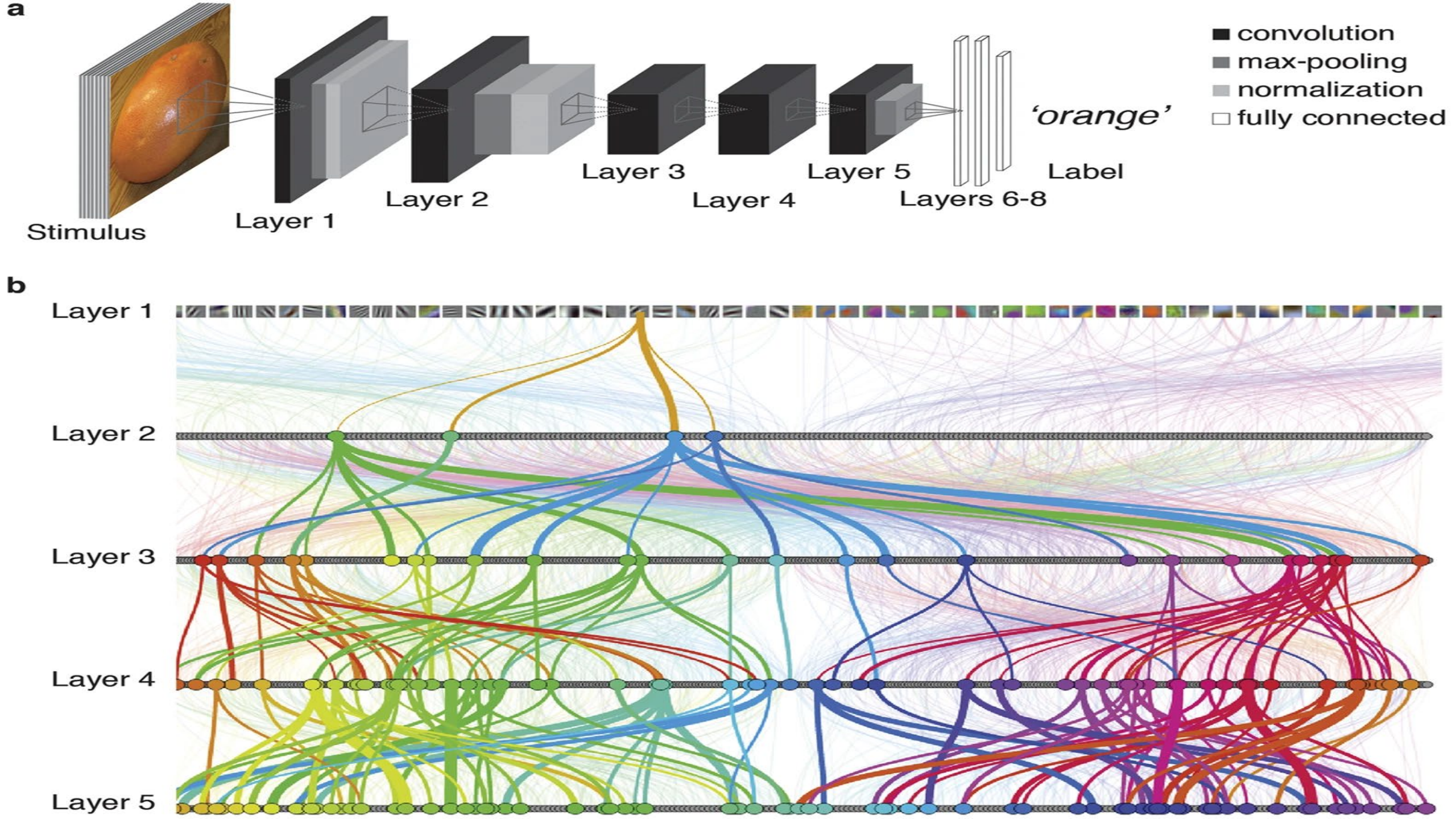


1. https://cacm.ac m.org/news/llm-hallucinations-a-bug-or-a-feature/

Faithfulness Hallucination	Instruction Inconsistency	*Instruction*: Translate the English question into Spanish: "What is the capital of France?" *Response*: The capital of France is Paris.	The user explicitly provided an instruction to translate the English question into Spanish. *Instead of performing the translation as instructed, the LLM chose to answer the question.*
	Context Inconsistency	*Retrieved context*: The Nile is a major north-flowing river in northeastern Africa. It's traditionally considered the longest river in the world. Originating in the Great Lakes region of central Africa, it flows through several countries before emptying into the Mediterranean Sea. *Instruction*: Summarize the retrieval document above. *Response*: The Nile originates from the mountain ranges of central Africa and flows into the Mediterranean.	The summary incorrectly states that the Nile originates from the "mountain ranges of central Africa", while the *paragraph mentions that the Nile originates in the Great Lakes region of central Africa.*
	Logical Inconsistency	*Instruction*: Solve the equation $2x + 3 = 11$ step by step. *Response*: Step 1: Subtract 3 from both sides to get: $2x = 8$. Step 2: Divide both sides by 2 to get: $x = 3$. So the result of the equation is $x = 3$.	The first step correctly isolates the term with x to get $2x = 8$. However, *the next step inaccurately divides 8 by 2 to yield a result of $x = 3$*, which is inconsistent with the earlier reasoning.

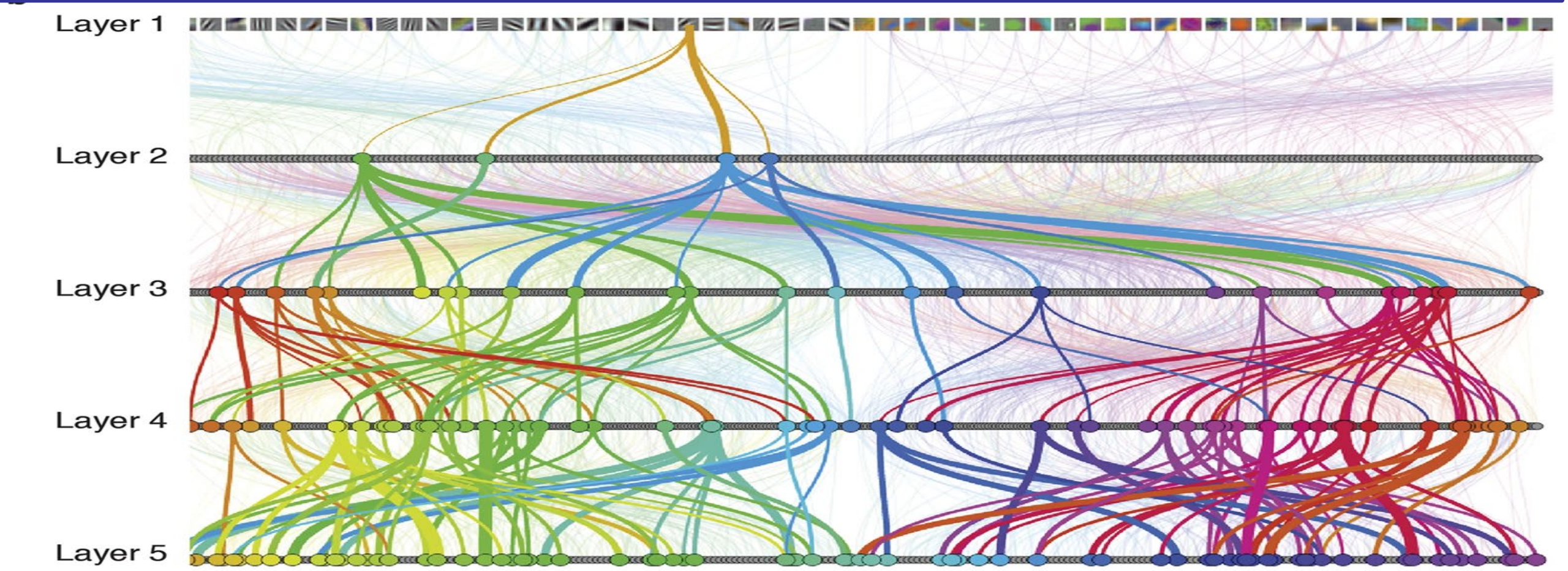
Generative AI Artwork: A Hallucination?




a
Stimulus
Layer 1
Layer 2
Layer 3
Layer 4
Layer 5
Layers 6-8
'orange'
Label
convolution
max-pooling
normalization
fully connected
b
Layer 1
Layer 2
Layer 3
Layer 4
Layer 5

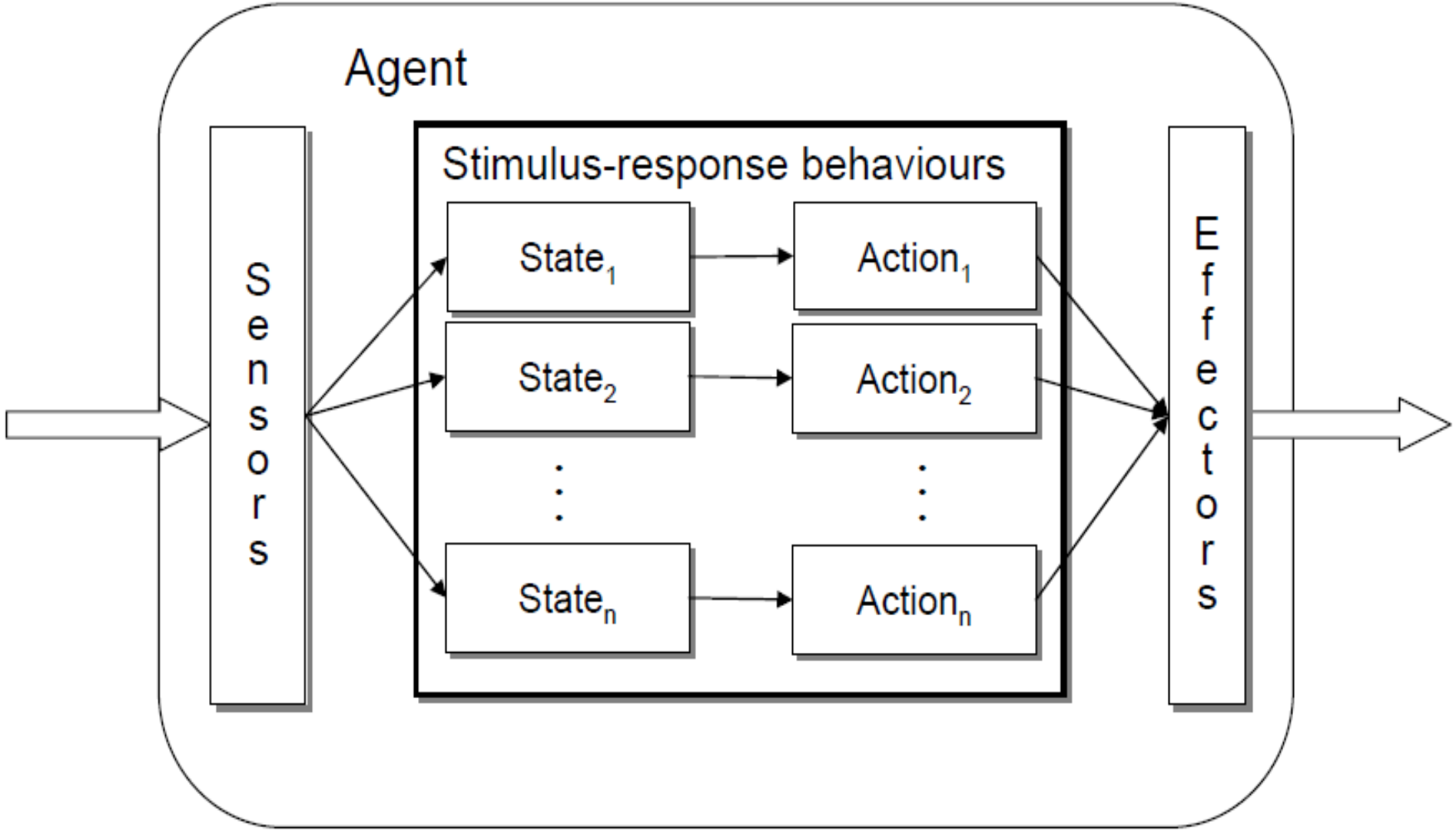
Hallucination Theory:

Failed generalisation from training data **(trained connection weights) to real-world data (previously unseen input) – Given enough scale, neurons (connection layers) interact in unpredictable ways.**



Is there a Simpler Approach?

Recall: Reactive Agents



Subsumption Architecture

Brooks' Thesis

- Intelligent behaviour can be generated without explicit representations (e.g., neural-based, symbolic-based).
- Intelligent behaviour can be generated without explicit reasoning (what neuro-, symbolic AI proposes).
- Intelligence is an *emergent property* of reactive components.

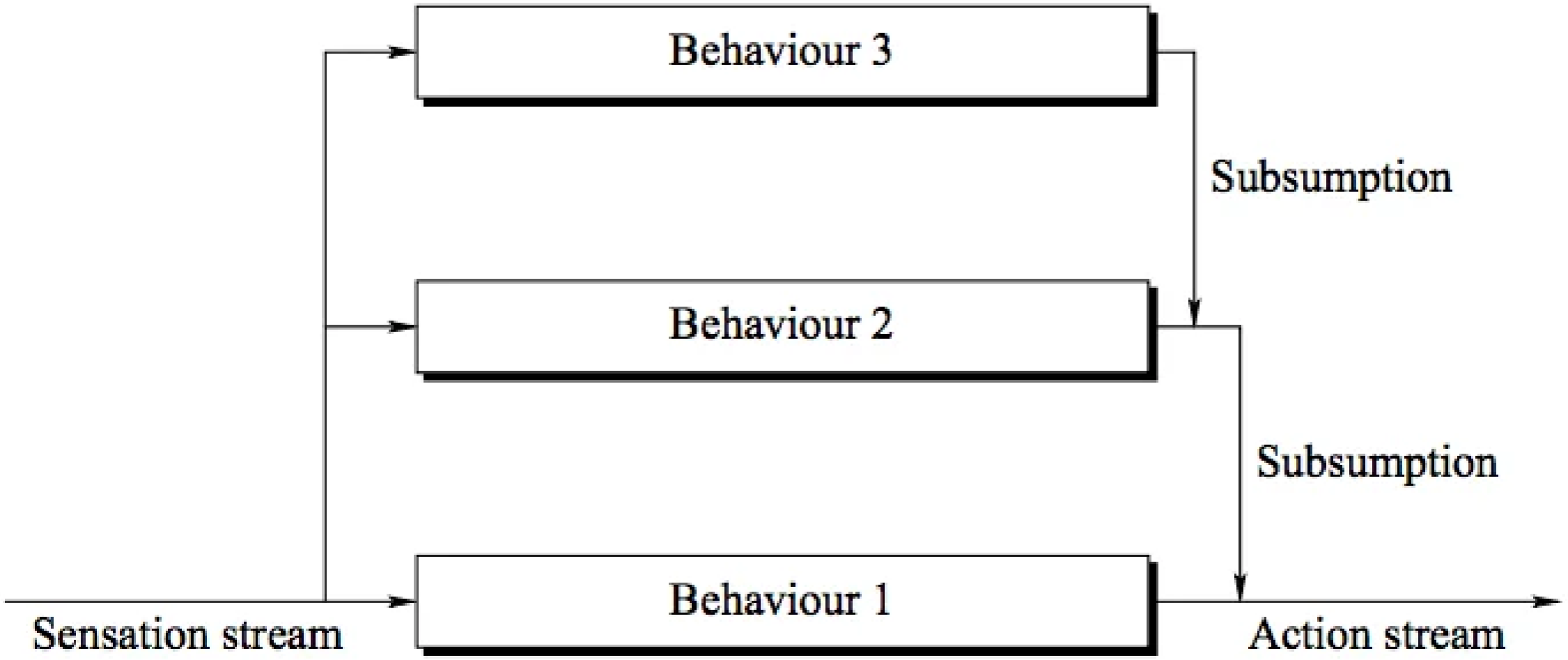
Subsumption Architecture

- “Real” intelligence is situated in the world – not in disembodied systems such as theorem provers, expert systems, LLMs.
- “Intelligent” behaviour arises as a result of an agent’s interaction with its environment.
- The subsumption architecture was designed for *embodied* (robotic) systems.

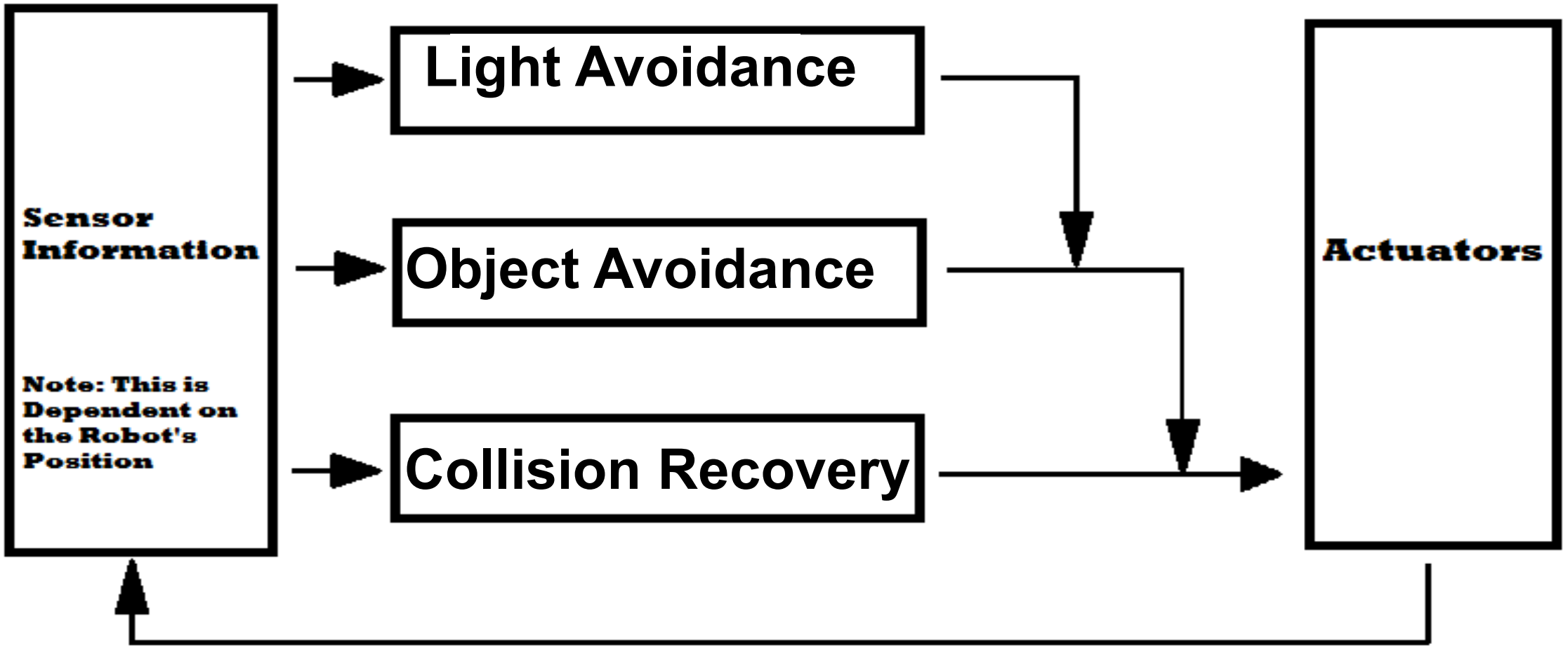
Subsumption Architecture

- A hierarchy of task-accomplishing behaviours.
- Each behaviour encodes simple rule-like structure.
- Lower layers represent more primitive kinds of behaviour (e.g., avoiding obstacles).
- Behaviour interaction in lower levels gives rise to higher level (task accomplishing) behaviours.

Subsumption Architecture



Light Avoiding Robot



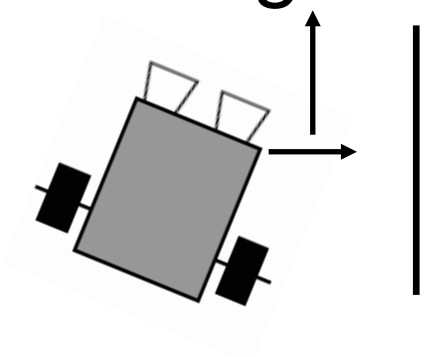


Maze Navigation

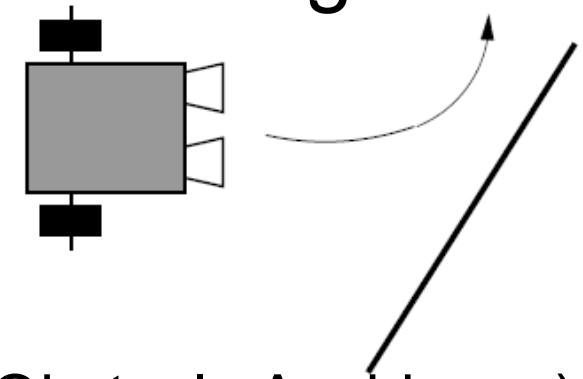
- Multiple (layered) behaviours:
  - **Collision Recovery –** If robot collides with an object, it backs up and turns about 90 degrees (in random direction).
  - **Object Avoidance –** Move forward but don't get too close (<5.0cm) to obstacles – turn about 90 degrees (in random direction).
  - **Explore World –** Move away from starting point.
- All behaviours run in parallel – Result is a maze navigation behaviour.

Maze Navigation

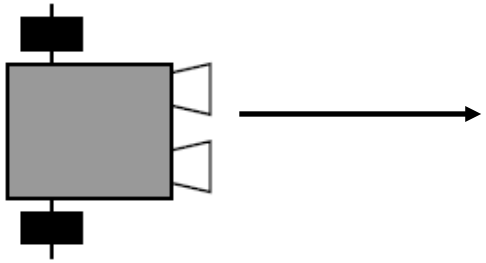
- Example – we code a robot with three simple behaviors:
  - Move forward (Obstacle avoidance).
  - Collision recovery.
  - Explore.
- Nothing in there about maze navigation.



Collision Recovery



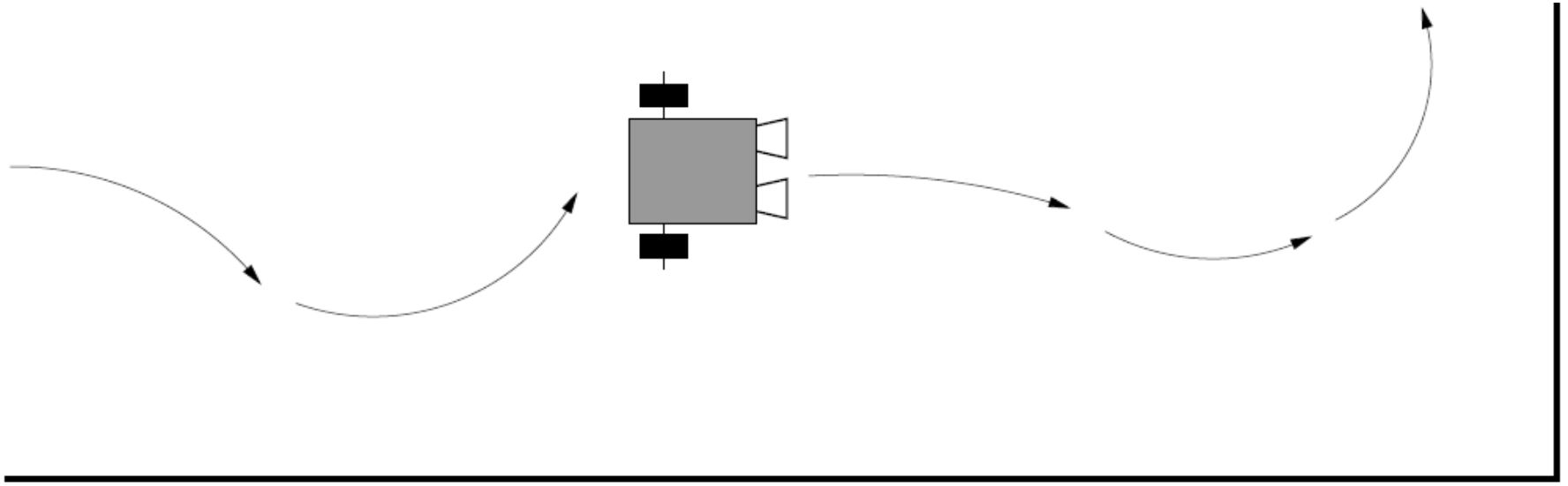
Move (Obstacle Avoidance)



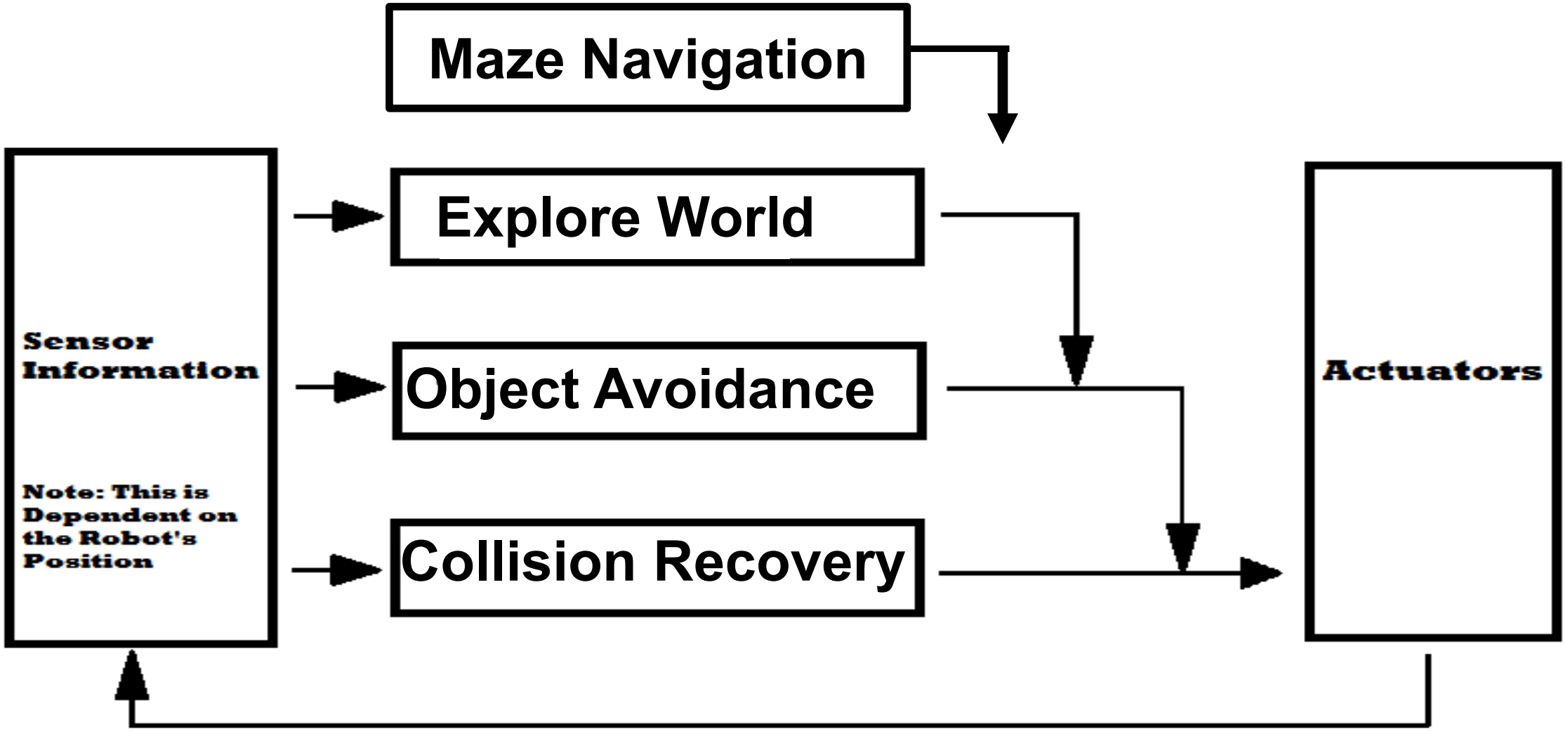
Explore

Maze Navigation is Emergent

- Robot itself is not aware of walls – only reacts to distance readings.
- Concepts: “wall”, “navigation” are not stored in the robot’s controller.
- The system is just a collection of behaviours (rules): *Collision recovery*, *Obstacle avoidance*, *Explore*.



Maze Navigation



Subsumption Architecture – Benefits

- **Simplicity, Modularity**: Each layer of behavior works independently – Makes the system easier to build, debug, and expand.
- **Responsiveness**: Sensory inputs without complex processing enable quick responses to environmental changes.
- **Scalability:** Basic behaviours added first, then more complex layers.
- **Fault Tolerance**: If one behaviour fails, other layers still work.

Subsumption Architecture – Problems?

Subsumption Architecture – Limits

- **Lacks Memory, Learning:** Without memory, robots cannot learn from experience, store complex data, limiting long-term adaptability.
- **Limited Planning:** Architecture's reactive nature isn't suited for tasks requiring detailed planning or sequential steps.
- **Difficult to Adapt to New Tasks:** Each behaviour (layer) is task-specific, making it hard to modify for new objectives.
- **No Abstract Concepts:** Without neural-, symbolic- processing, it cannot handle tasks involving language or abstract reasoning.

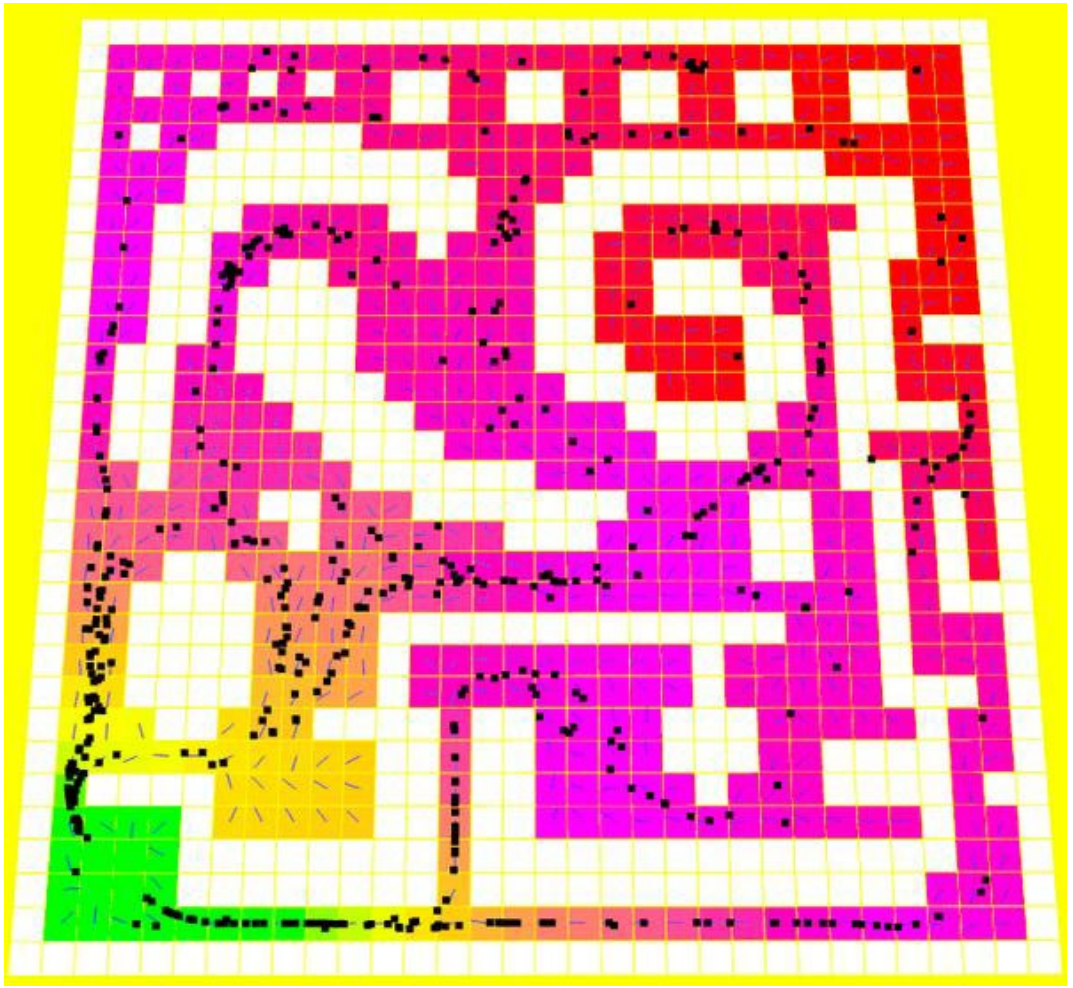
Other Reactive Architectures

- Multi-agent system models → Emergent behaviour.
  - Boids[1]
  - **Continuum crowds[2]**
  - Stigmergy model[3]

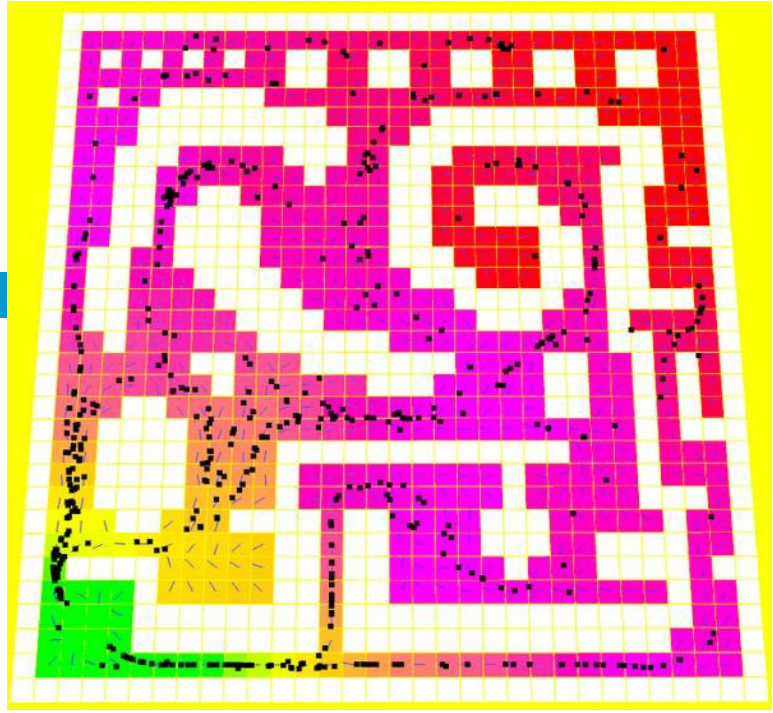
1. Reynolds, C. (1987). Flocks, herds and schools: A distributed behavioral model. Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.
2. Treuille, A., et al. (2006). Continuum crowds. ACM Transactions on Graphics, 25(3): 1160 – 1168.
3. Theraulaz, G., & Bonabeau, E. (1995). Coordination in Distributed Building. Science 269: 686-688.

Continuum Crowds

- Used in CGI applications (e.g., games, movies).
- Heat-map:
  - **Red:** Area of highest discomfort.
  - **Purple:** High discomfort.
  - **Orange**: Low discomfort.
  - **Green:** Area of lowest discomfort.
- Agents follow four simple rules.



Continuum Crowds: Rules



- Each agent tries to reach a geographic goal **(G)**.
- Agents move at the maximum speed possible **(f)**.
- There is a *discomfort field* **g**, so agents prefer to be at point **x** rather than **x0**, if **g(x0) > g(x)**.
- Let **P** be the set of all paths from **x** to some point in the goal **(G)**:
  - Assuming speed **f**, discomfort **g**, goal **G** are fixed, agent at location **x** will pick the path **p ∈ P** minimizing: *path length*, *time*, *discomfort*.

Other Reactive Architectures

- *Subsumption*.
- **Boids**[1]
- *Continuum Crowds*.
- Stigmergy.

1. Reynolds, C. (1987). Flocks, herds and schools: A distributed behavioral model. Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.

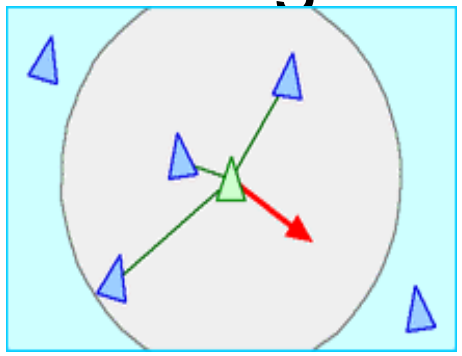
Flocks, Herds and Schools

- **Boids flocking model**[1] – To visualize flocking behaviour for computer graphics applications.
- **Reactive (MAS) model:** Each agent is a rule-set (no internal state!).
- Flocking behaviour emerges from N local agent interactions.
- Type of emergent (flocking) behaviour can be changed via changing parameter values in agent rules.

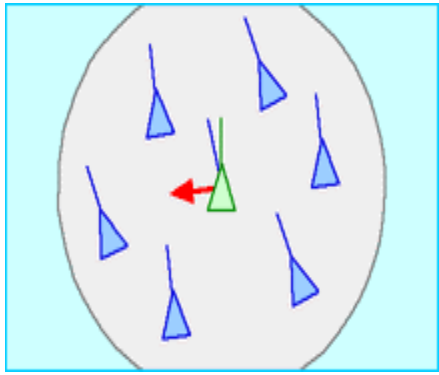
1. Reynolds, C. (1987). "Flocks, herds and schools: A distributed behavioral model". Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.

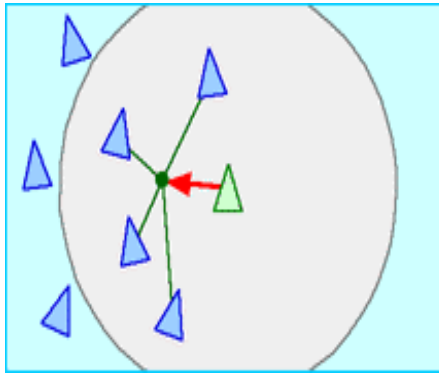
Boids: Flocking

- Agent (*Boid*) reacts only to flock mates in its neighbourhood.



- **3 Steering rules:**
  - **Repulsion:** Steer to avoid local flock mates.
  - **Attraction:** Go towards average heading of local flock mates.
  - **Orientation:** Go towards average position of local flock mates.





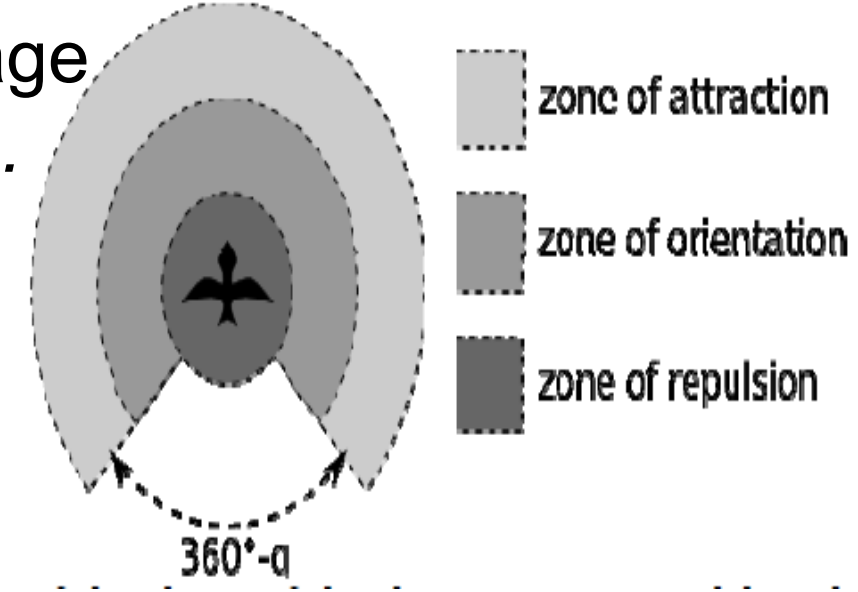
Boids Rules

- r_c (repulsion) ≤ r_o (orientation) ≤ r_a (attraction)

- **Rule 1 (Repulsion):** Move away from agents in circle (with radius r_c) choosing direction opposite to average direction of agents in circle.

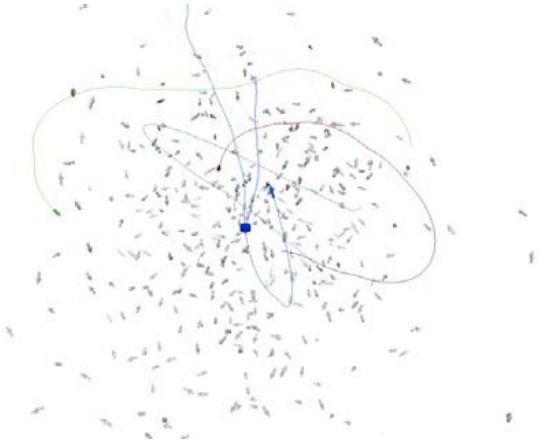
- **Rule 2 (Orientation):** Agent is oriented in average direction of other agents between radii r_o and r_a .

- **Rule 3 (Attraction):** Agent moves in average direction of all agents between radii r_c and r_a .

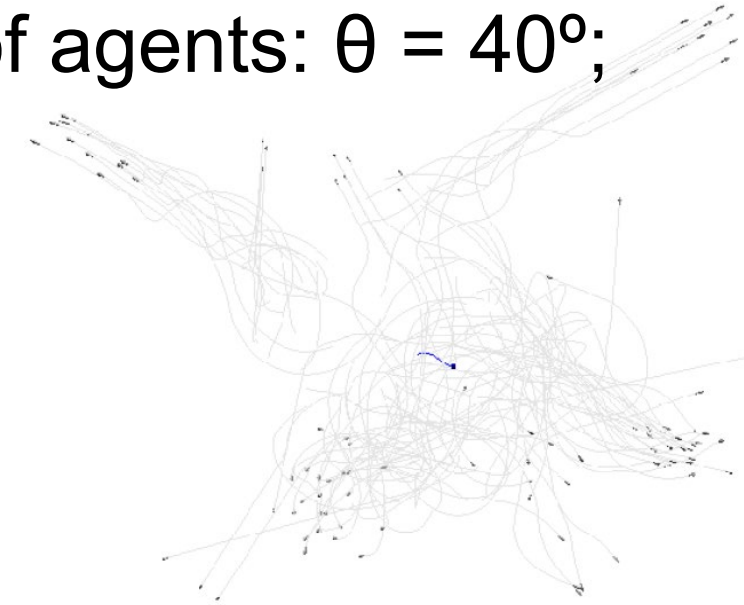


Boids (Emergent) Behaviours

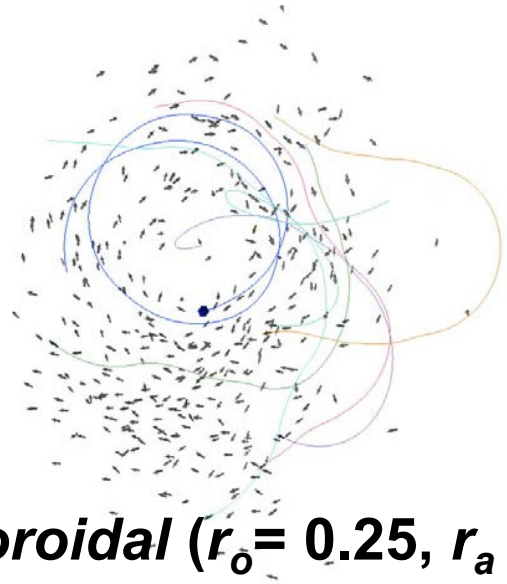
- Choose parameter values: $r_r \leq r_o \leq r_a$
  - View angle: α = 270º ;
  - Average direction of agents: θ = 40º;
  - Agents: N = 400.



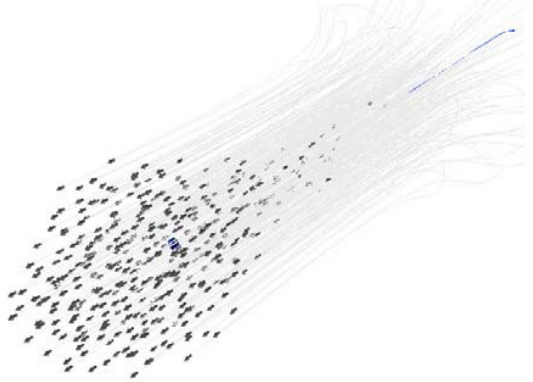
Swarming **(r_o = 0.1, r_a = 0.85)**



Dispersion **(r_o= 0.15, r_a = 0.25)**

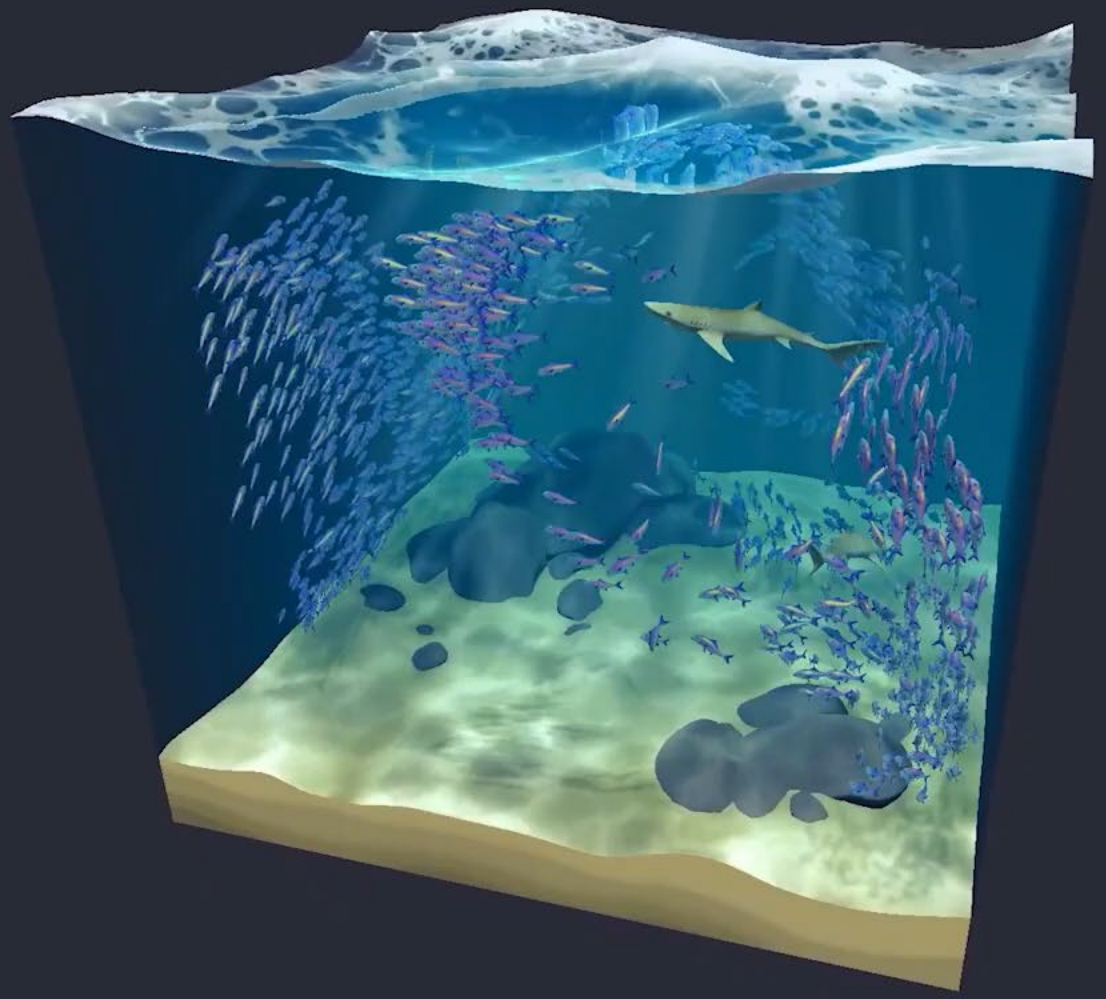


Toroidal **(r_o= 0.25, r_a = 0.85)**



Meteor **(r_o= 0.45, r_a = 0.85)**

Combining the Boids and Continuum Models…



Reactive Architectures – Any Problems?

Reactive Architectures – Problems

- How to get a desired (collective) behaviour from N agent behaviours is not obvious – Impractical for complex task solving.
- Subsumption architecture:
  - How many layers?
  - What rules (behaviours) to encode?
  - What will the emergent behaviour be?
- Boids:
  - Can the Boids solve (real-world) tasks?

Summary: Types of Architectures

- **Reactive:** Agent maps sensor data to actions – Decisions based on environmental stimuli rather than memory or predictive capabilities.
- **Deliberative:** Agent makes decisions based on reasoning, planning. Internal world model used to predict future outcomes.
- **Cognitive:** Agent uses ANN/DL (Connectionist) models (to try) to mimic human-like, reasoning, learning and decision-making:
  - Black-box approach, (usually) supervised and reinforcement learning.
- **Hybrid:** Combining *reactive*, *deliberative* and *cognitive* elements.

What you should Know

- Difference between reactive, deliberative and hybrid agents.
- The subsumption architecture – what it is, its benefits and limits.

Reading (On Amathuba!)

- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* – 4th Edition – Ch2 (Intelligent Agents): aima.cs.berkeley.edu/
- Poole, D., & Mackworth, A. (2023). *Artificial Intelligence: Foundations of Computational Agents* – Ch1:https://artint.info/3e/html/ArtInt3e.Ch1.html
- Brooks, R. (1991), Intelligence Without Representation, *Artificial Intelligence*, 47(1991): 139-159.
- Newell, A., & Simon, H. (1976). Computer science as empirical inquiry: Symbols and search. Communications of the ACM, 19(3): 113–126.
- Wei, J., et al. (2022). Emergent abilities of large language models. arXiv:2206.07682.
- Harnad, S. (1990). The symbol grounding problem. Physica D: Nonlinear Phenomena, 42(1-3): 335–346.
- Gahrn-Andersen, R. (2025). Beyond symbol processing: the embodied limits of LLMs and the gap between AI and human cognition. *AI and Society*, 40: 3105–3107.

